



FULL STACK DEVELOPER

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HELLO!

At this stage of the interview, you will work on a project that reflects the kind of problems we solve at Panteon every day. This isn't just a coding exercise — it's a chance for us to see how you think, how you approach a problem, and how you work with AI tools to build something real.

GENERAL INFORMATION

You are responsible for designing a leaderboard system that integrates into an existing backend stack. The system runs on **Node.js**, **PostgreSQL**, **MongoDB** and **Redis**. Your implementation must stay within this stack.

Client and server code should be in separate projects. You are expected to use **TypeScript** for both front end and back end.

SCENARIO

One of Panteon's idle/clicker mobile games has grown faster than expected. The game has over **10 million registered players** and sees around **2 million active players every day**. Players earn in-game currency as they play, and each week starts fresh — whoever earns the most that week climbs to the top of the leaderboard.

The original leaderboard was built quickly and has been running since launch. It works — but barely. The backend team gets weekly complaints from players:

"The leaderboard takes forever to load."

"I can see the top players fine, but I can't find my own rank."

"My friend is in the top 50 but the page just freezes when I scroll down."

The product team's ask is simple: "Make the leaderboard instant. Players should see their own rank and the players around them. Rewards should go out automatically at the end of the week."

You are expected to design and build this system from scratch.

REQUIREMENTS

System Requirements

- Architecture of the system should be stateless
- Your application will be tested on both PC and mobile
- Client and server code should be in separate projects

Requirement from the Designer

The weekly leaderboard system is built around automatically collecting 2% of the total money players earn during the week into a prize pool. At the end of the week, this pool is distributed to the top 100 players according to ranking: 1st place gets 20%, 2nd gets 15%, 3rd gets 10%; the remaining 55% is distributed among players ranked 4th through 100th, based on their rank. Once prize distribution is complete, both the pool and the leaderboard reset to start the new week.

On the leaderboard screen, players should always see the ranking of the top 100. If a player is not in the top 100, they should still be able to see their own position so they don't lose track — in that case, along with their own rank, the 3 players above and the 2 players below them should also appear in the list.

The Interface

We want you to design a leaderboard screen for competitive players. The moment a player opens it, they should clearly see their own ranking, easily compare themselves with others, and have a smooth experience. We expect you to think about how you'll handle the interactions that make the list easier to explore, the opportunities for global comparison, and the weekly reward/status communication. Don't forget to include sample data so we can test it — any creative touches beyond that are yours to make.

CRITERIAS

Technical:

- Suitability with the scenario and requirements
- Preferred technologies and their appropriate use
- Scalability
- Performance
- Code quality
- Cloud usage
- Reusable React components

AI-Assisted Development:

We use AI tools heavily in our daily workflow — and we expect you to as well. What matters to us is **how** you use AI, not just whether you use it. There's no required format: use whatever tools feel natural to you (Claude Code, Cursor, Copilot, ChatGPT — anything goes). Just make sure to include something in your project that shows us your AI workflow: which tools you used, where they helped, and where you made your own calls. A single markdown file, inline comments, a short README section — it's up to you.

SAVE & SEND!

You are expected to share your project via BitBucket or GitHub. Send us an email that contains the link to the project — hr@panteon.games

- A working production build should be deployed on an accessible domain
- If you can't understand the requirements perfectly, feel free to ask

G O O D L U C K A N D H A V E F U N !

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